Chapter 16

Complex multiple fixed and combined fixed and removable prosthodontics

David Bartlett, David Ricketts

CHAPTER CONTENTS

Indications **201**
Combined fixed and removable prosthodontics **201**
  Rest seats, guide planes and undercuts 202
  Precision attachments 203
    Patient 1 203
    Patient 2 204
    Patient 3 205
  Shortened dental arch 205
Multiple fixed prostheses **205**
Management of tooth wear **206**
  Restorative management 207
    Surgical crown lengthening 207
    Overdentures 208
    Creating occlusal clearance 208
Summary **213**

INDICATIONS

Complex multiple fixed and combined fixed and removable prosthodontics can be indicated to restore teeth damaged and depleted by the ravages of caries, tooth wear and inherited and acquired defects (outlined in Chapters 1–5).

COMBINED FIXED AND REMOVABLE PROSTHODONTICS

Many partial dentures are made using support from teeth restored with existing plastic restorations. Designing crowns and partial dentures at the same time helps to improve the retention, support and stability of the denture. The denture must be designed before crowns are made to ensure the rest seats, guide planes and undercuts are placed in the optimum position. A well-designed denture will need upper and lower study casts and ideally these should be mounted on an articulator. This can be an average value articulator which gives an indication of the intercuspal position and allows some degree of excursive movements, or for more complex cases involving multiple crowns, a semi-adjustable articulator should be used to more accurately allow assessment of functional movements of the mandible. The study casts should be surveyed to show the position of undercuts on the teeth or where undercuts are required in crowns. This process follows conventional prosthodontic procedures and is covered by the appropriate texts. Making crowns with guide planes, undercuts and rest seats increases their complexity and cost. Therefore, the design of tooth support, retention and stability for the partial denture needs to be carefully planned to be optimal and to ensure cost effectiveness.

© 2011 Elsevier Ltd.

Rest seats, guide planes and undercuts

Rest seats provide direct dental support for a partial denture. They need to be large enough to provide support and deep enough for the rest on the partial denture to fit into the cavity or rest seat in the crown (or tooth) without interfering with the occlusion. The size of the rest seats needs to be sufficient to provide support and normally about one-third of the proximal width of the tooth is sufficient. The shape of the cavity should avoid sharp internal line angles and follow the general shape as shown in Figure 16.1. There should also be sufficient occlusal clearance to allow a minor connection between the base of the partial denture and any rest seats and clasps around the same tooth (Figure 16.2).

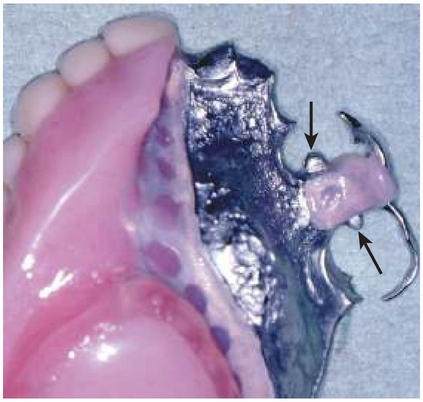

Figure 16.1 The fit surface of this partial removable cobalt chromium prosthesis shows the general outline of the rests (arrowed) having a rounded character in all dimensions. Note the obturator bottom left which seals a maxillary defect created following surgery to remove a tumour.

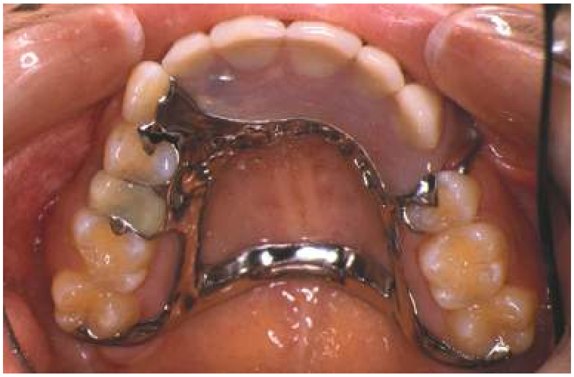

Figure 16.2 Sufficient occlusal clearance is required not only for the rest seat, but also the connector between the rest seat (and clasps if present) and base of denture.

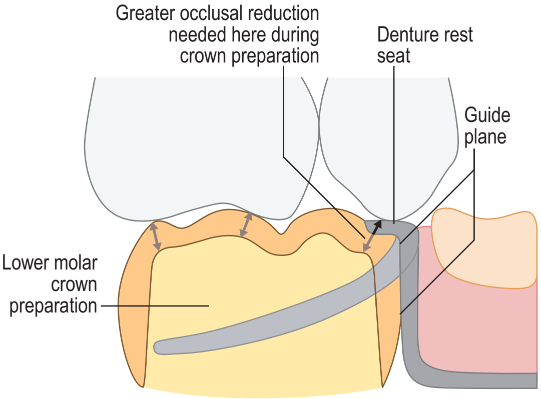
Greater occlusal reduction needed here during crown preparation
Denture rest seat
Guide plane
Lower molar crown preparation

Figure 16.3 During crown preparation, the occlusal reduction in the region where a rest seat is planned has to be greater to allow for the thickness of the rest on the denture and the metal of the crown.

Knowing where the rest seat and connectors will ultimately be placed will therefore influence tooth preparation when a crown is planned. Occlusal reduction in the region of the rest seat and connector has to be greater than elsewhere on the occlusal surface as sufficient interocclusal space has to be provided for both the metal substructure to the crown and the thickness of the rest itself (Figure 16.3). Where crowns are concerned it is always important to place rest seats into metal as placement into ceramic will lead to shear fractures (Figure 16.4).

The retention of metal-based partial dentures relies heavily upon clasps. To create the necessary resistance to displacement of the clasps, an undercut, normally on the buccal surface of the crown (tooth), needs to be present. When designing crowns and partial dentures the depth of undercut required for cast cobalt chromium clasps is

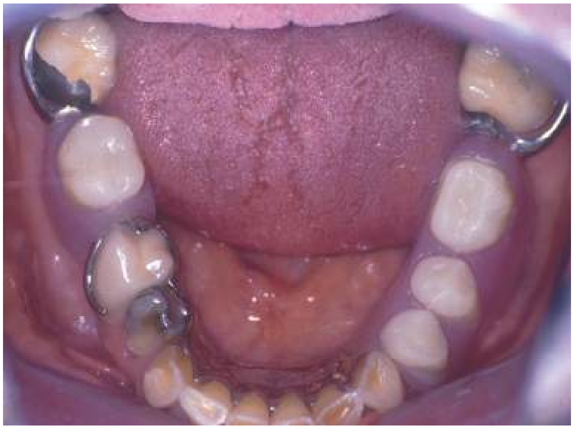

Figure 16.4 Inappropriate placement of rest seats onto ceramic in the metal–ceramic crowns on the lower molar teeth has led to ceramic fracture.

generally 0.25 mm; to a degree, however, this will depend on the length of the clasp arm (longer clasp arms can engage deeper undercuts). For wrought clasps a deeper undercut can be engaged (0.5 mm) as these have greater inherent flexibility. These undercuts have to be prescribed to the technician so that they can be incorporated into the crown contour.

Guide planes designed on the proximal surfaces of crowns increase the surface area in contact with the partial denture saddle area and increase the resistance to displacement and so retention. The guide planes, when present, need to be parallel to each other to maximize the retention. There is little value in having a single crown with a guide plane without reciprocating action from another crown or tooth.

This section emphasizes the importance of planning the denture design before any tooth preparation takes place. Figures 16.5 and 16.6 illustrate this point clearly.

Precision attachments

There are situations where precision attachments are still indicated when fixed and removable prosthodontics are considered together. The routine use of implants has, to some extent, reduced the need for precision attachments on teeth because the cost of the attachment and laboratory work can approach that of implant retained prostheses. Precision attachments linked to fixed laboratory-made prostheses can be used to retain and stabilize removable prostheses. There are a number of designs that can be considered, ranging from extracoronal stud or seeker (ball and socket) attachments to intracoronal movable joints similar to those used in conjunction with fixed–movable bridge designs (see Chapter 19). To illustrate how such precision attachments can be used, consider the following three clinical scenarios.

Patient 1

The patient seen in Figure 16.7 has missing lower posterior teeth with severe resorption of the edentulous ridge, making it difficult to tolerate a lower removable prosthesis due to the lack of retention and stability. To address this, the lower canine teeth have been prepared for metal–ceramic crowns which have distal cantilevered bars and stud attachments incorporated into the metal substructure (Figure 16.8). The lower removable cobalt chromium prosthesis has a corresponding female attachment (socket) embedded into the acrylic of the denture base (Figure 16.9). This 'ball and socket' arrangement provides sufficient retention and stability without the need for clasping anterior teeth (Figure 16.8).

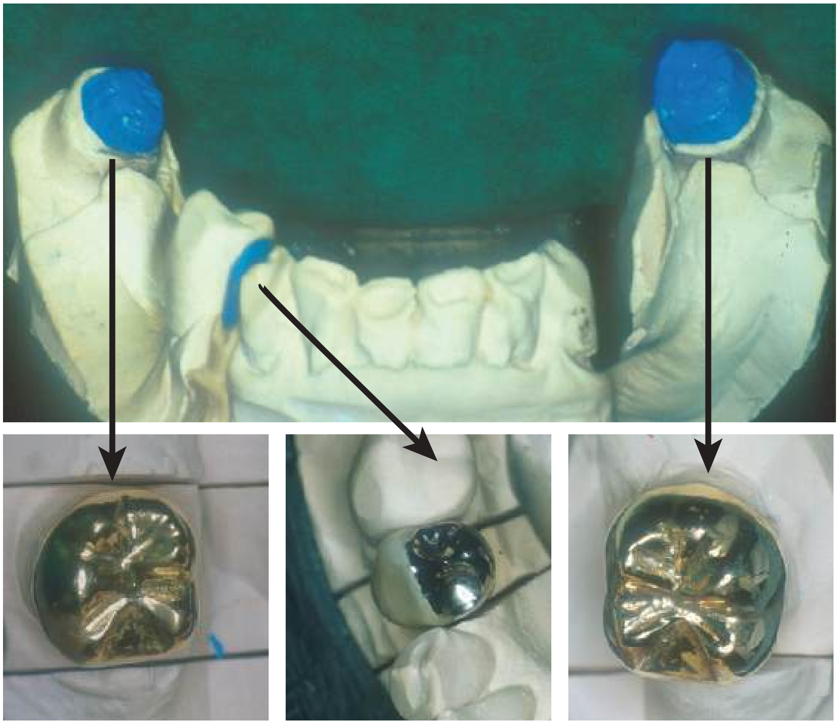

Figure 16.5 Having a denture design prior to preparation of the lower molar teeth and lower right premolar tooth for crowns has enabled appropriate prescription of rest seats, undercuts and guide planes in the indirect restorations.

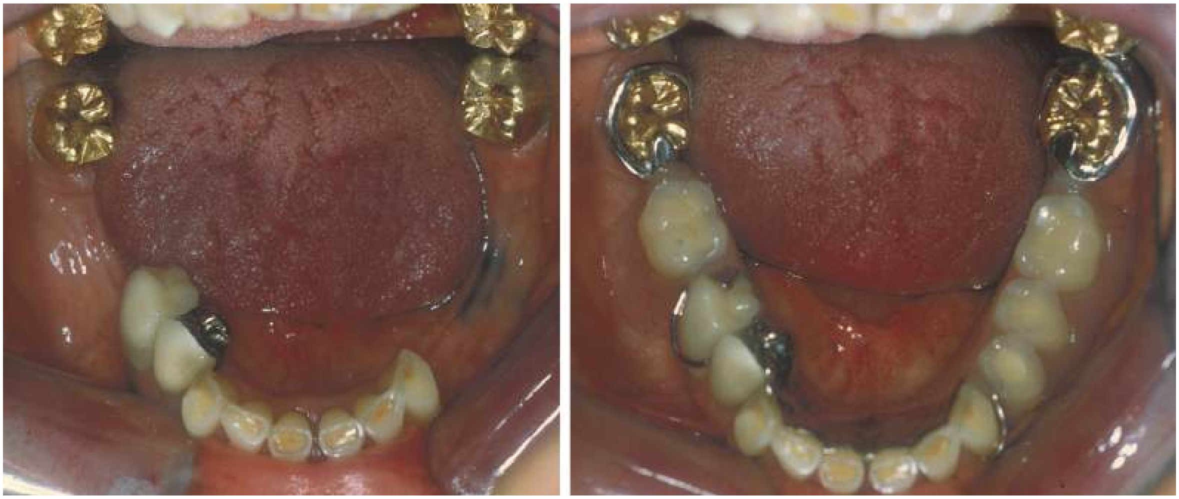

Figure 16.6 The crowns for the patient seen in Figure 16.5 cemented (left) and the new cobalt chromium denture (right).

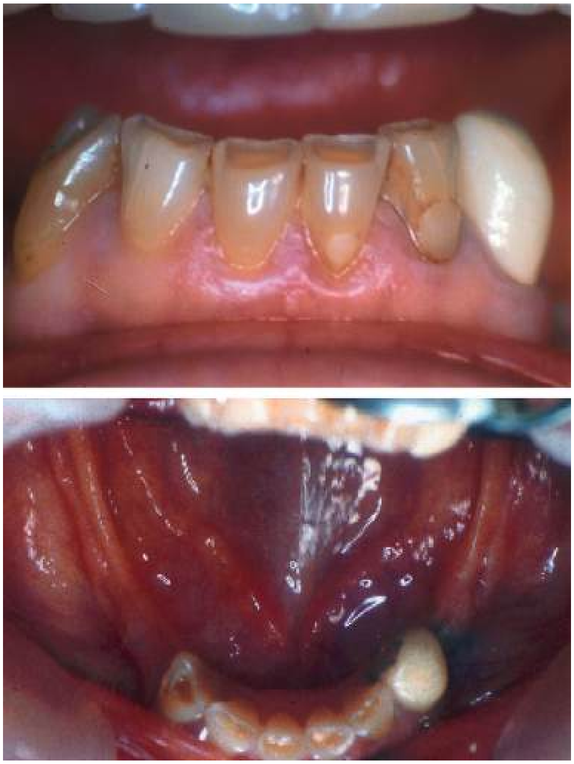

Figure 16.7 Patient with lower anterior teeth only. Due to the severe resorption of the lower edentulous ridge the patient is unable to tolerate their existing denture due to poor retention and stability.

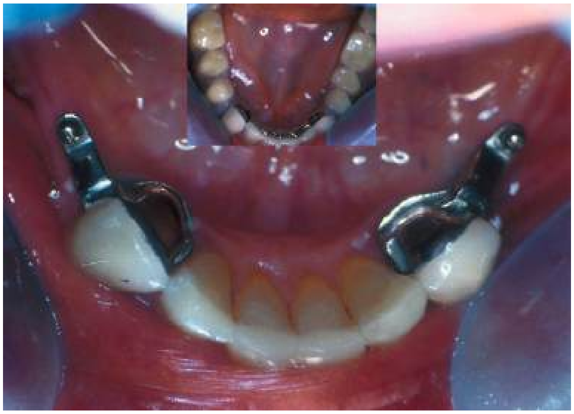

Figure 16.8 The patient seen in Figure 16.7 with crowns on the canine teeth that incorporate cingulum rest seats and distal extension bars with a stud attachment. The denture (inset) therefore avoids unsightly clasps in the anterior region.

Patient 2

Precision attachments can be used in conjunction with other conventional means of retention for removable prostheses. Consider the patient seen in Figure 16.10; gold crowns have been placed on the molar teeth incorporating rest seats, guide planes and undercuts to achieve support, stability and conventional retention; however, retention is not so easily achieved anteriorly without the presence of unsightly metal clasps. One way to overcome this is to use a precision attachment on the upper left

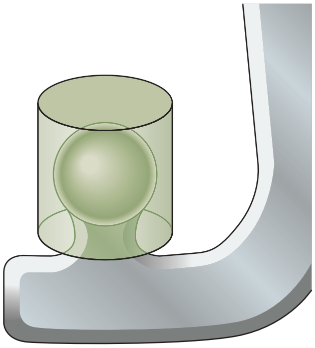

Figure 16.9 Diagrammatic representation of stud attachment on the distal cantilever arm. The metal housing for the socket joint is embedded into the denture base. The socket joint is provided by plastic colour-coded inserts of varying stiffness and hence retentiveness.

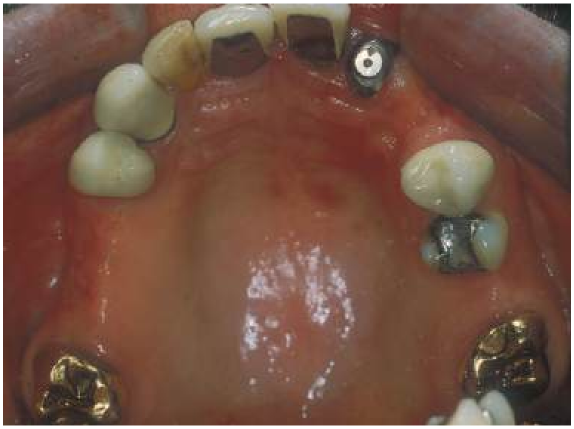

Figure 16.10 Patient for which an upper cobalt chromium prosthesis is planned. Gold crowns have been placed on the two molar teeth incorporating rest seats, guide planes and appropriate undercuts. Additional retention is gained anteriorly with a Rothermann-type precision attachment incorporated into a cast post and diaphragm on the left lateral incisor.

lateral incisor root. This tooth has been root treated and prepared for a cast post and diaphragm onto which is soldered the male component of a Rothermann type precision attachment; the female 'clip' attachment is embedded into the fit surface of the denture (Figure 16.11). The male component consists of a disc with a concavity around its periphery; the female clip seats into this concavity on insertion of the denture, giving good retention.

Patient 3

Figure 16.12 shows the die stone model of a patient who presented with multiple retained roots in the upper arch; the only teeth with natural crowns were the central incisors and the upper left second molar tooth. He was unable to wear a conventional denture due to a marked retch reflex. As for the previous patient the roots were root filled and prepared for cast post, diaphragm and Rothermann type attachments (Figure 16.13). The precision attachments provide excellent retention and allow a horseshoe cobalt chromium denture to be made with minimal palatal coverage.

With the advent of dental implants there is greater scope to retain, stabilize and support removable prostheses in edentulous patients. Studs, bars and magnets can be attached to implants with reciprocal components bedded into the denture base. Implant retained removable prostheses are outwith the scope of this text and the reader should refer to an appropriate source of information.

Shortened dental arch

The concept of the shortened dental arch is discussed in further detail in Chapter 17; however, of relevance to this section, it should be re-emphasized that not all missing teeth need to be replaced. If posterior teeth are extracted and the change in oral function is minor, there is little need for tooth replacement providing a stable occlusion is maintained. If the teeth are visible in the smile line, arguably there is a need to replace the teeth. The other important consideration when assessing the need to replace teeth is cost. From a health economic perspective anterior teeth and the first premolars provide adequate function for most of the population. Admittedly, this ignores the aesthetic and psychological components of replacing posterior teeth. Retention, stability and support for free-end saddles often results in poor compliance for these dentures, resulting in an unstable restoration compared to teeth.

MULTIPLE FIXED PROSTHESES

Most patients need crowns or bridges on only a few teeth. However, when patients need more extensive use of crowns and bridges, or a combination of fixed and removable prostheses that affect most of the occlusal table, careful planning is needed. For example, patients requiring extensive treatment are those suffering from severe tooth wear. When most or all of the teeth need restoration, a change to the intercuspal position can be planned, often at an increased occlusal vertical dimension due to short clinical crowns and the need to avoid further occlusal reduction. When this is done a *reorganized approach* to the occlusion is said to have been adopted and some refer to this as a 'full arch rehabilitation'. This is in contrast to when indirect restorations

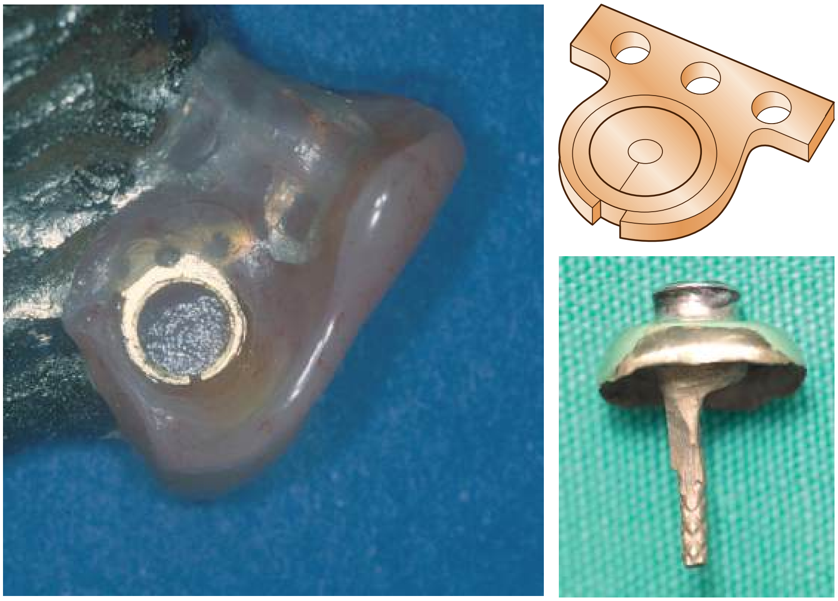

Figure 16.11 Female component of Rothermann-type attachment embedded in the denture base (left), the male component attached to the cast post and diaphragm (bottom right) and diagrammatic representation of how the female component clips into the recess on the sides of the male component (top right).

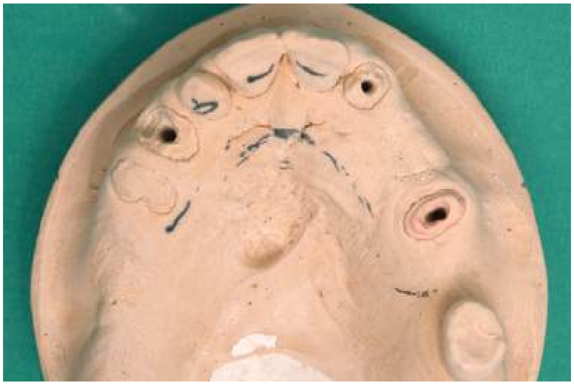

Figure 16.12 Model of patient whose retained roots have been root treated, and prepared for cast posts and diaphragms onto which precision attachments can be attached.

are provided to fit in harmony with the exiting intercuspal position, the so-called *conformative approach*. Restoring teeth adopting a conformative approach is sufficient for the majority of situations, as usually only smaller numbers of teeth are restored. The conformative approach is a simpler procedure whereas a full-mouth rehabilitation and reorganized approach will almost certainly need a semi-adjustable articulator and careful treatment planning and execution.

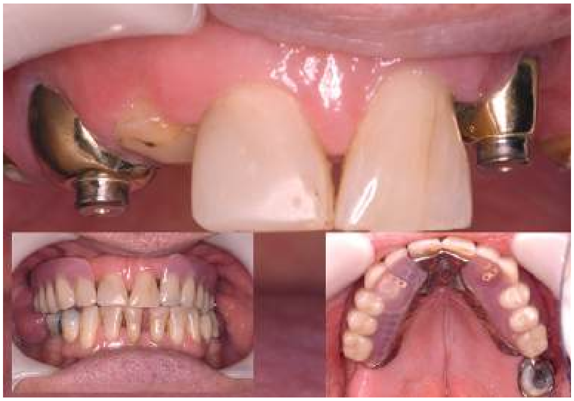

Figure 16.13 Patient whose working model is seen in Figure 16.12. The cast posts, diaphragms and precision attachments cemented (top) and horseshoe cobalt chromium partial denture fitted. The precision attachments provide adequate retention, support and stability to the upper partial prosthesis.

MANAGEMENT OF TOOTH WEAR

There is no justification to the concept that tooth wear inevitably leads to the total destruction of teeth. In some individuals the extent of the tooth wear may compromise the longevity of the tooth; however, in most patients, once prevention has been instituted, the process continues slowly and is part of the ageing process. For many patients monitoring of tooth wear is an effective and acceptable procedure, even though there is no attempt to restore the shape and appearance of the teeth. For the preventive management of tooth wear, see Chapter 4. The restoration of worn teeth is normally expensive and can require specialist levels of care, mainly because it rarely affects single teeth and more commonly involves quadrants or even whole dental arches.

Restorative management

The main indications for restorative management are to improve the appearance of teeth and to prevent intractable sensitivity (Figures 16.14 and 16.15). In some cases restorations are indicted to prevent wear compromising the survival of the tooth. Once the decision to restore teeth has been taken, the clinician needs to assess how much tooth height is present and whether or not there is sufficient vertical space for the restorations. The critical decision is whether there is sufficient proximal height of the teeth to allow conventional crown preparations, bearing in mind that additional occlusal reduction may be needed to make space for a crown (Figure 16.14).

Surgical crown lengthening

Crown preparations on already shortened teeth due to tooth wear may result in an unacceptable retention. If insufficient tooth height remains, crown lengthening will be needed to reposition the gingival margin in a more apical direction. In this procedure a muco-gingival flap is raised, crestal alveolar bone is removed and thinned slightly, and the flap replaced in a more apical position. Surgical access makes the procedure easier to provide on the buccal surfaces of upper anterior teeth and more difficult lingually and palatally. The palatal mucosa is also more firmly attached to the underlying bone, increasing the difficulty in reflecting the tissue to gain access to the alveolar bone. Normally, crown lengthening of upper anterior teeth is possible because of their relatively long roots but lower incisors have shorter roots and are more challenging (Figures 16.16 and 16.17).

Crown lengthening is not without its disadvantages. Whilst a longer clinical crown is achieved, interocclusal clearance is not, and further occlusal or incisal reduction is required, often necessitating elective root canal treatment (Figure 16.17). Once crestal alveolar bone is removed, bone support is lost and an adverse crown:root ratio may be produced. Additionally, as the longitudinal shape of roots, particularly of anterior teeth, is tapered, the more apically the gingival margin is placed, the larger the interdental

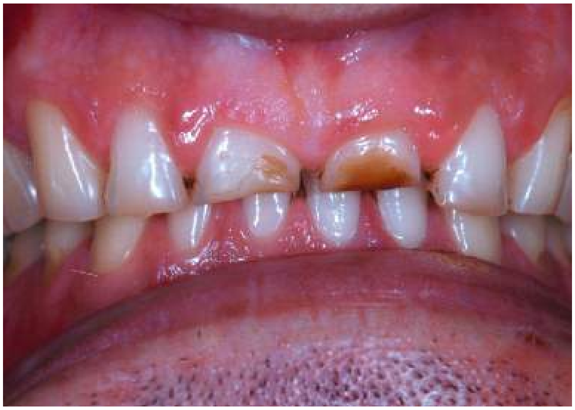

Figure 16.14 Extensively worn upper central incisors leading to a poor appearance. The mesial and distal height of both teeth is insufficient for conventionally prepared crowns. During crown preparation approximately 1–2 mm of occlusal and incisal reduction will be required to create space for the metal and porcelain if a metal–ceramic crown is chosen. In this case further reduction of the tooth would produce unretentive preparations.

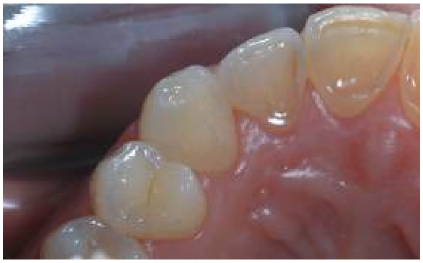

Figure 16.15 Intractable sensitivity in relation to tooth wear is rare but can justify restorations, particularly if the pulp might be compromised if delayed any further.

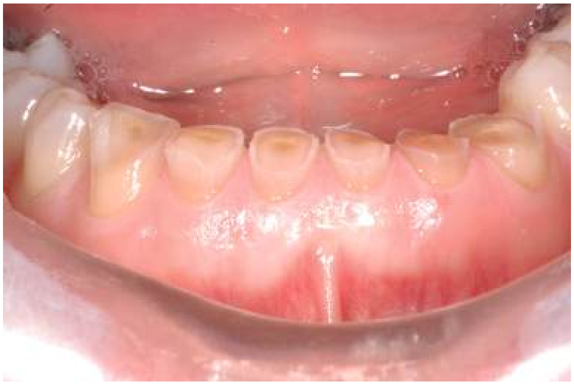

Figure 16.16 These teeth are too short for conventional restorations and need crown lengthening.

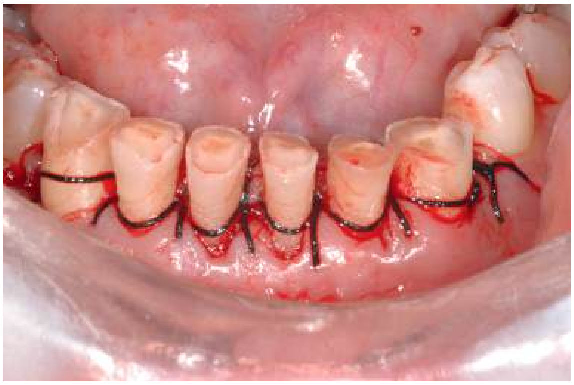

Figure 16.17 The results after crown lengthening for the patient seen in Figure 16.16. To place crowns on these teeth will require incisal reduction, which is likely to necessitate elective root canal treatment as the reactionary (tertiary) dentine is already visible.

space becomes (Figure 16.18) and crowns are either produced with poor emergence profiles to overcome this or large interdental black triangles have to be accepted. Finally, surgical crown lengthening is an uncomfortable procedure that some patients may find unacceptable.

Overdentures

If the clinical crown height and root length are too short for crown lengthening, then either the teeth can be preserved as overdenture abutments (Figure 16.19) or extracted and replaced by dentures or implants.

Creating occlusal clearance

In patients with tooth wear that requires operative intervention, how they are managed will, to a large extent, depend on whether the tooth wear is localized or generalized and how heavily restored the remaining dentition is.

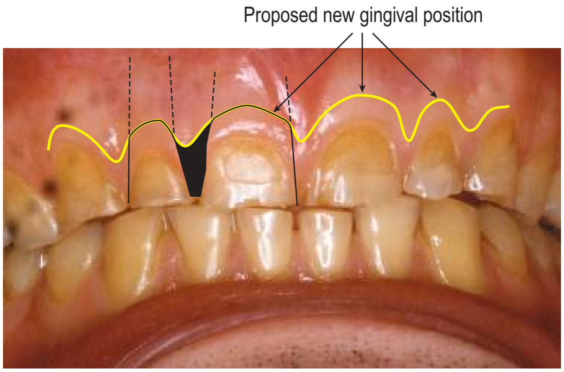

Figure 16.18 Crown lengthening in the upper anterior region can lead to unsightly interdental black triangles due to the tapered roots (outlined in broken lines).

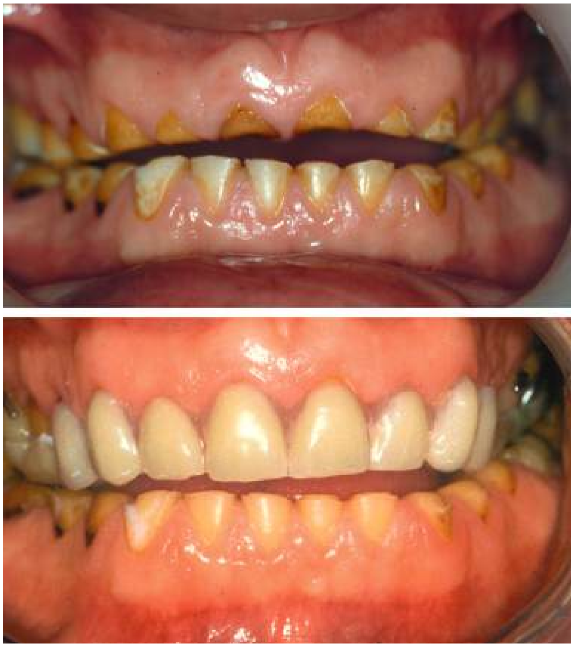

Figure 16.19 Severely worn teeth (top) where crown lengthening was not considered and restored with an overdenture (bottom).

Localized tooth wear

Localized tooth wear is often seen confined to the upper and/or lower anterior teeth. Despite the amount of wear the teeth are commonly in occlusal contact in intercuspal position; this is due to dento-alveolar compensation (Figure 16.20). If the worn teeth are in occlusal contact there is insufficient space to place restorations; however, occlusal (incisal) reduction to achieve interocclusal space would foreshorten the teeth even more and to a level that crowning would be impossible. Interocclusal space can be created either with conventional orthodontic intrusion/extrusion using fixed appliances or with a bite plane – the Dahl concept.

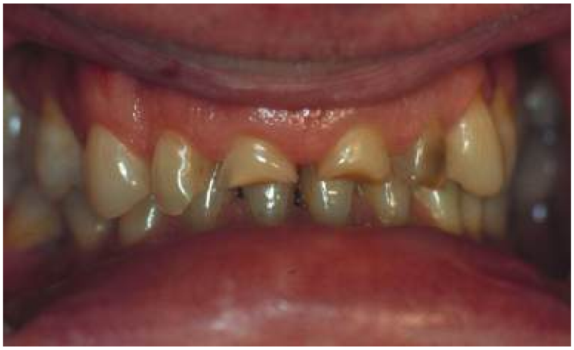

Figure 16.20 Patient with severe tooth wear of the upper anterior teeth; however, there is no interocclusal space in intercuspal position due to dento-alveolar compensation.

The Dahl concept utilizes intrusive forces and the eruptive potential of teeth to create differential movement between groups of teeth, so creating sufficient occlusal vertical space necessary for a crown without any significant occlusal reduction. The Dahl concept has received considerable attention and been shown to be a reliable and effective management for short clinical crowns. In preserving occlusal/incisal tooth tissue the added benefit can be preservation of pulpal health which would otherwise be compromised if additional tooth reduction was necessary.

Early Dahl appliances were made using partial dentures or metal occlusal splints which were placed between the worn teeth (Figures 16.21 and 16.22). The anterior plane encourages the intrusion of the opposing teeth and as the splint prevents contact of the remaining teeth it allows their overeruption. Research by the Dahl group showed that the initial tooth movement was mainly intrusion of the teeth opposing the splint, followed by overeruption of the teeth propped apart. Commonly at the end of treatment overeruption exceeded intrusion which resulted in a small increased vertical dimension. Usually sufficient interocclusal space is created in 3–4 months (Figure 16.23).

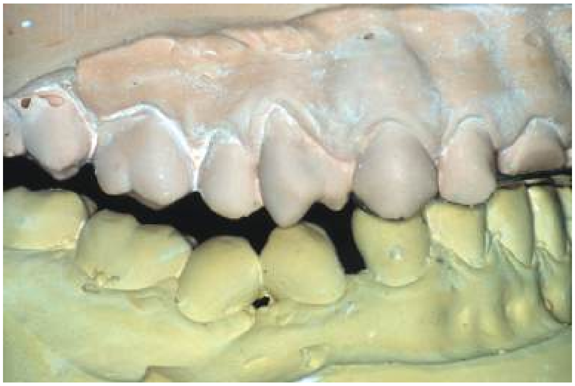

Figure 16.21 Construction of a fixed metal Dahl appliance for the anterior teeth of the patient seen in Figure 16.20. Note the degree of separation of the posterior teeth.

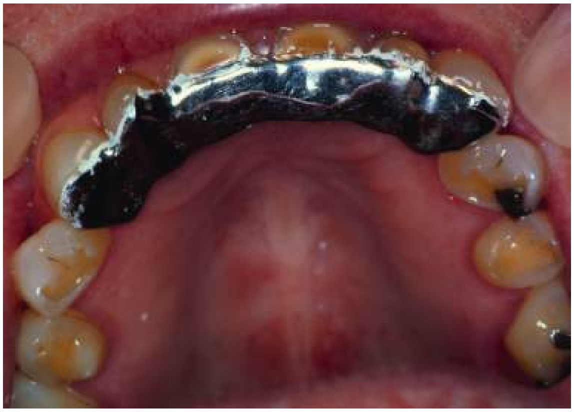

Figure 16.22 Patient seen in Figure 16.20 with the fixed metal Dahl appliance cemented to the upper anterior teeth.

In the experience of clinicians using this technique, precipitation of symptoms of mandibular dysfunction does not occur. However, the presence of symptoms before treatment normally contraindicates the use of the Dahl principle. A common clinical finding with patients suffering from severe tooth wear is good periodontal health; however, poor oral hygiene or increased periodontal probing depths over 3 mm also contraindicate the use of this technique.

Whilst early work on Dahl appliances concentrated on removable or fixed bite planes, most clinicians today use composite build-ups shaped to the ideal contours of the teeth to create the space (Figures 16.24 and 16.25).

The Dahl movement is only applicable when the tooth wear is localized to a few teeth. The differential movement of the teeth produces the space necessary to crown the worn teeth and removes the need to remove additional tooth tissue to provide occlusal clearance. If the tooth wear is generalized the movement is ineffective and a full-mouth rehabilitation is indicated.

Stages in the treatment of localized tooth wear

Articulated study casts mounted on a semi-adjustable articulator using a retruded contact position interocclusal record will provide the clinician with an opportunity to plan the patient's care. The probable reason why Dahl movements are successful is that patients with tooth wear resulting in short clinical crowns and no interocclusal space have a proven dento-alveolar adaptive capacity. Therefore, they are more likely to respond favourably and without symptoms if the height of the teeth is restored to their original dimensions. Articulated study

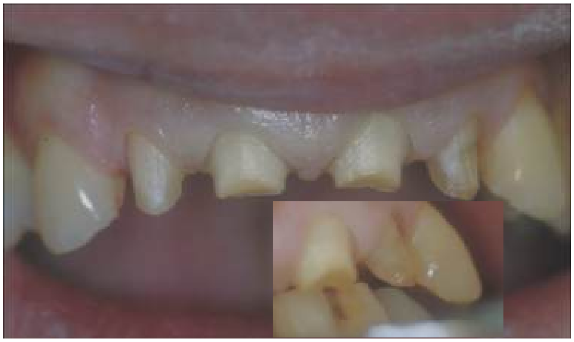

Figure 16.23 Patient seen in Figure 16.20. Preparation of the upper incisors for crowns has been possible without incisal or palatal reduction as the Dahl appliance has created sufficient differential tooth movement and interocclusal space (inset). The canines were restored with palatal gold veneers (see Chapter 12).

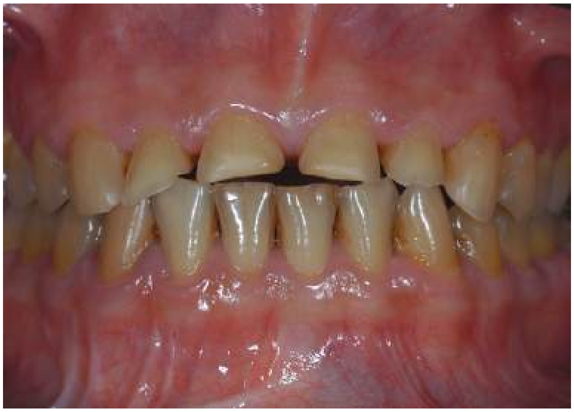

Figure 16.24 Patient with tooth wear and dento-alveolar compensation. Most Dahl movements are now created by restoring the teeth to their ideal shape and appearance by using composites.

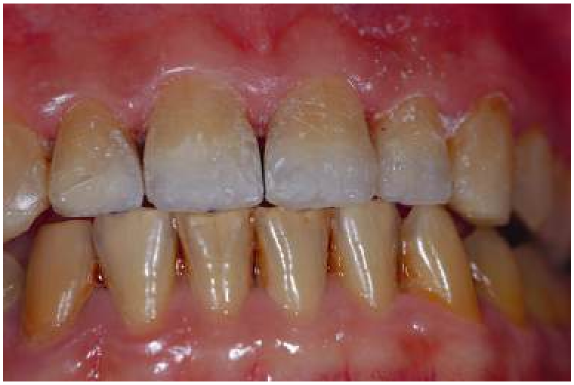

Figure 16.25 The composite Dahl appliance for the patient seen in Figure 16.24. Note the posterior teeth are separated to allow overeruption.

casts allow the technician and clinician to plan the tooth shape in wax and determine how much interocclusal clearance is required. This diagnostic process is driven by the ideal shape and height of teeth (see Chapter 15). Previous photographs and other visual aids may provide some idea of the original tooth dimension; however, if not, then the shape of the tooth needs to be formed using basic principles. There will obviously be individual variations but the diagnostic wax-ups should be made to conform to the shape and size of the patient's teeth.

The next phase of treatment will depend on the clinician's personal preference. Although some practitioners still use a fixed Dahl appliance, most prefer to restore the teeth with composite prior to crowns (Figures 16.24 and 16.25). The advantage of using composite, particularly in the anterior part of the mouth, is that the build-ups act as the Dahl appliance and also act as a provisional restoration. If the diagnostic wax-up has not fully predicted the ideal shape and size of the teeth, an intraoral adjustment of the restorations creates the optimum appearance. Once built up, the composites need to have an even occlusal contact with the opposing teeth (Figure 16.26). The adjustment is critical to the survival of the restorations. If, after the composite additions, only a few teeth have occlusal contacts, there is an increased risk of fracture of the restoration as they become overloaded and are more likely to cause discomfort to the patient than is otherwise necessary. There is reasonable evidence to suggest that composites used to restore worn anterior teeth will survive between 3 and 5 years (Gow & Hemmings, 2002; Hemmings et al., 2000; Redman et al., 2003). During that time a certain amount of adjustment with composite additions may be necessary to preserve the restorations.

At some point the composites will need replacing and this is often managed using crowns. In some patients the transition to crowns occurs more quickly, particularly if repeated fractures of the composite occur. Once the decision to restore the teeth with crowns is made, either the original diagnostic wax-up can be reused or a new one can be created, or indeed the existing composites can be used to plan the shape of the indirect restorations. A custom-formed incisal guidance table as described in Chapter 14 can be used to ensure the anterior guidance is reproduced if necessary.

A fundamental indicator to the success of restoration of worn teeth is the amount of tooth structure remaining. The Dahl movement only creates interocclusal space but if the height of the natural teeth is too short there will be insufficient retention of crowns. Following crown preparation any residual composite remaining on the tooth cannot be considered reliable retention for crowns in the long term (Figure 16.27). Although composite additions give the illusion of the ideal shape and height of a crown

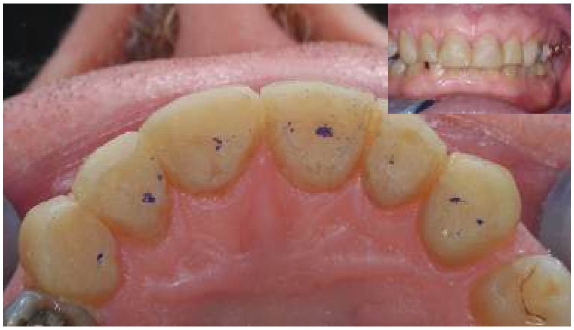

Figure 16.26 Composites to build up worn teeth. The composites were initially used as a Dahl appliance and even occlusal contact on all the Dahl teeth is essential. Inset: posterior tooth contact achieved.

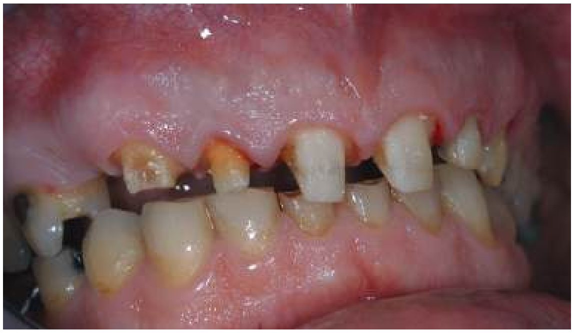

Figure 16.27 The tips of the two central incisors have composite additions. The bond between composite and tooth cannot be relied upon over the long term.

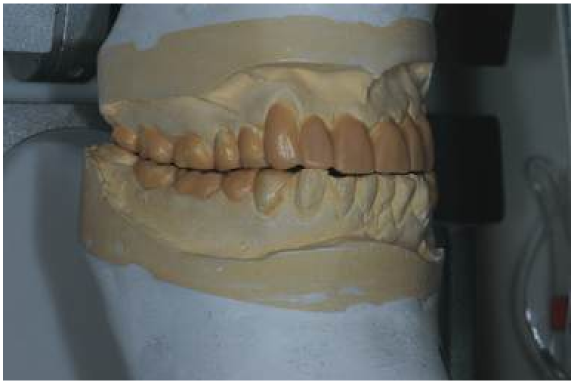

Figure 16.28 A diagnostic wax-up of a patient requiring multiple crowns. The shape and the height of the teeth are created not only from the basic shape of the standing teeth but also using the width and height ratio.

preparation, their bond to tooth is insufficient over the lifetime of the crown.

There is a mounting evidence base to support the use of Dahl movements to restore worn teeth. The technique is not universally accepted and there is scepticism, particularly in North America. However, many other parts of the world have adopted the technique. The work by Gough and Setchell (1999) suggests that the time needed for the adaption and occlusal changes ranges from 3 to 9 months. The time increases when the incisal relationship is a Class II Division II and is less successful with Class III and Class II Division I relationships. If insufficient teeth meet in the intercuspal position, a Dahl movement is unlikely to be successful and more radical intervention is necessary.

Generalized tooth wear

Where the tooth wear is generalized and affects most teeth the differential movement using a Dahl restoration is inappropriate. In these more severe cases the distribution of tooth wear is rarely restricted to one arch or one quadrant. Often there is a mixed outcome, with some severely worn teeth in one arch opposed by less worn teeth in the other. The restoration of the mouth should be aimed to increase the height of the worn teeth but to leave, where possible, the less severely worn without restorations.

The procedure for restoring a dentition with an increased number of restorations follows a similar procedure to the work-up needed for the Dahl appliance. Upper and lower study casts mounted on a semi-adjustable articulator using a retruded contact position interocclusal record provide the opportunity to plan the treatment. The diagnostic wax-up (Figure 16.28) allows the height and shape of the teeth to be predicted and to determine the increased occlusal vertical dimension that is required. Provisional restorations can then be made based on the wax-up, either in composite directly built up in the mouth or from provisional crowns. The provisional crowns can be made from a vacuum-formed splint made on a duplicate model of the wax-up. These provisional restorations can then be used to assess the appearance, occlusion and function over a period of time. An individual assessment of the available tooth structure of each tooth is required and an estimate made of the effect of tooth preparation for a crown. If that assessment predicts there will be insufficient tooth remaining to provide adequate retention, crown lengthening could be considered. Once the provisional restorations are fitted, further occlusal adjustment is needed to ensure the shape conforms to the patient's expectations and that there is even occlusal contact. Provided the patient and dentist are happy, the definitive crowns can be made based on the shape and contour of the provisional restorations.

When a full arch rehabilitation is considered and a reorganized approach is adopted to manage the occlusion, restorations should be made using a terminal hinge axis position at a set occlusal vertical dimension. The terminal hinge axis is used as it is the most reproducible jaw relationship and it facilitates alteration of the occlusal vertical dimension (see Chapter 6). The most demanding part of making crowns for a full arch rehabilitation is the need to ensure even occlusal contact and smooth guidance.

In a full mouth rehabilitation at an increased occlusal vertical dimension the procedure can either be planned according to groups of teeth or teeth in one arch can be prepared and restored all at once. The latter is demanding and extremely time consuming for the dentist, and arduous for the patient. A quadrant or sextant approach overcomes this problem and can be carried out in two ways:

- Teeth can be prepared in groups at different appointments with two lots of provisional restorations being made, the first set at the original vertical dimension and conforming to the existing occlusion and the second at the reorganized occlusion at an increased vertical dimension. The first set will be used until all preparations are carried out and at this stage the second set will be used to assess the increased occlusal vertical dimension and occlusion according to the diagnostic wax-up. A single impression can then be taken of all the prepared teeth and all restorations made on one model. This has the benefit that all the characteristics and shade of tooth-coloured restorations will match. All of the restorations will then be fitted at the subsequent appointment, ensuring even occlusal contacts throughout to prevent any overeruption of teeth. This process is dependent upon accurately articulated study casts. Once all the interocclusal contacts are removed, the demands on the technician making the crowns and the dentist fitting them increase significantly.

- Teeth can be prepared in groups, an impression taken and then the provisional or definitive restorations made in groups. If the definitive restorations are cemented, the occlusal clearance created for the remaining arch has to be maintained. This can be with composite build-ups on the remaining teeth which have to be placed at the same appointment the crowns are cemented or more simply by using an occlusal splint which is adjusted accordingly. Consider the patient in Figure 16.29: the teeth in the upper arch are worn and the posterior

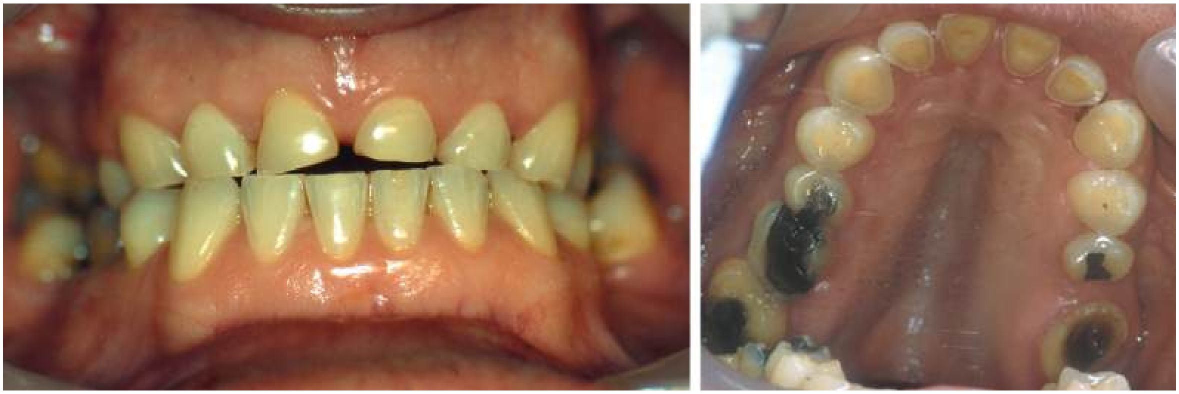

Figure 16.29 Patient with marked tooth wear and heavily restored posterior teeth for whom a full upper arch rehabilitation is planned.

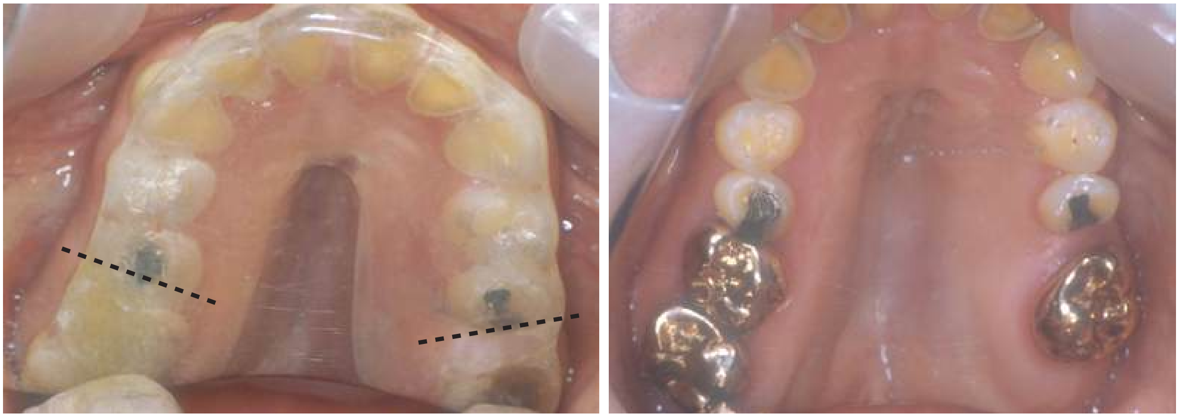

Figure 16.30 Patient seen in Figure 16.29 with Michigan splint at the desired increased occlusal vertical dimension (left). Once the gold crowns and onlays are cemented on the molar teeth (right), the splint can be cut back to the dashed lines. Wear of the splint over the premolar, canine and incisor teeth will maintain the occlusal clearance created. Even occlusal contact on crowns and splint must be ensured.

teeth are heavily restored. An upper full arch rehabilitation is planned. A Michigan splint is made to assess the increased occlusal vertical dimension planned (Figure 16.30). The molar teeth are prepared for onlays and crowns at the increased vertical dimension. The crowns are made and tried in to ensure bilateral occlusal contact and cemented. The splint is then cut away from over the new gold crowns so that it sits over the premolar and anterior teeth only, maintaining the occlusal clearance achieved over these teeth (Figure 16.31). These teeth can then be prepared at a subsequent appointment for composite onlays (premolar teeth) and dentine-bonded crowns (anterior teeth) (Figure 16.31).

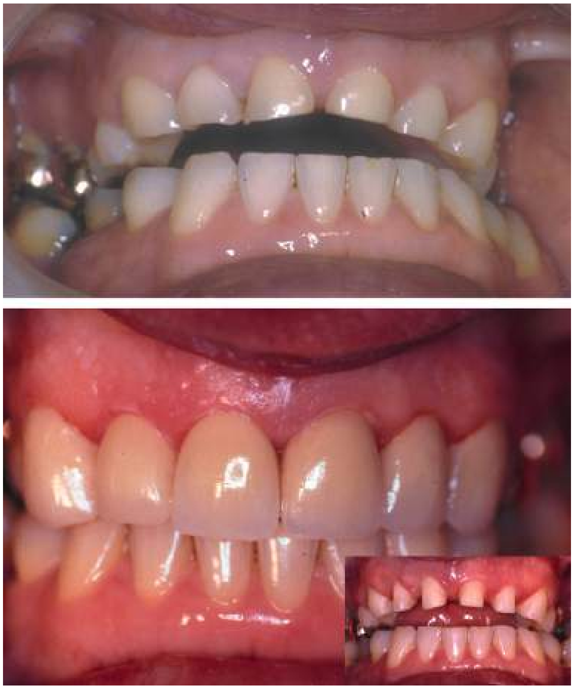

Figure 16.31 Patient seen in Figures 16.29 and 16.30. The Michigan splint retained anteriorly has maintained the occlusal clearance created by the posterior gold restorations, allowing sufficient space for dentine-bonded crowns.

SUMMARY

The restoration of dentitions with multiple fixed or a combination of fixed and removable prostheses is complex and requires careful decision making and planning. It also comes with a cost in both monetary terms and maintenance of oral health in the long term. Such complex dentistry should not be considered lightly and should only be undertaken with the full informed consent of the patient.

FURTHER READING

Gough, M.B., Setchell, D., 1999. A retrospective study of 50 treatments using an appliance to produce localised occlusal space by relative axial tooth movement. Br. Dent. J. 187, 134–139.

Gow, A.M., Hemmings, K.W., 2002. The treatment of localised anterior tooth wear with indirect Artglass restorations at an increased occlusal vertical dimension. Results after two years. Eur. J. Prosthodont. Restor. Dent. 10, 101–105.

Hemmings, K.W., Darbar, U.R., Vaughan, S., 2000. Tooth wear treated with direct composite restorations at an increased vertical dimension: results at 30 months. J. Prosthet. Dent. 83, 287–293.

Redman, C.D., Hemmings, K.W., Good, J.A., 2003. The survival and clinical performance of resin-based composite restorations used to treat localised anterior tooth wear. Br. Dent. J. 194, 566–572.

Ricketts, D.N., Smith, B.G., 1993a. Minor axial tooth movement in preparation for fixed prostheses. Eur. J. Prosthodont. Restor. Dent. 1, 145–149.

Ricketts, D.N., Smith, B.G., 1993b. Clinical techniques for producing and monitoring minor axial tooth movement. Eur. J. Prosthodont. Restor. Dent. 2, 5–9.